

EXHIBIT 14



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013
133964	7590	08/27/2018		
HOFFMAN WARNICK LLC			EXAMINER	
540 Broadway			NGUYEN, LAWRENCE-LIN T	
4th Floor				
ALBANY, NY 12207			ART UNIT	PAPER NUMBER
			2897	
			NOTIFICATION DATE	DELIVERY MODE
			08/27/2018	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

PTOCommunications@hoffmanwarnick.com

Office Action Summary	Application No. 15/826,939	Applicant(s) ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN	Art Unit 2897	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 11/30/2017.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a) ☐ This action is **FINAL**. 2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1-6 is/are pending in the application.
5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☒ Claim(s) 1-6 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement.

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☐ The drawing(s) filed on _____ is/are: a) ☐ accepted or b) ☐ objected to by the Examiner.
Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).
Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. _____.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)
Paper No(s)/Mail Date 11/30/2017.

3) ☐ Interview Summary (PTO-413)
Paper No(s)/Mail Date. _____.

4) ☐ Other: _____.

Detailed Action

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.
2. This non-final Office action is responsive to Applicants' application filed **04/06/2018**. Claims **1-6** are presented for examination and are rejected for the reasons indicated herein below.
3. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS*. See MPEP 2141.02 VI.

Claim Rejections - 35 USC § 102(a)(2)

4. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

A person shall be entitled to a patent unless –

(a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

5. Claims **1-6** are rejected under 35 U.S.C. 102(a2), as being anticipated by **Alptekin et al., (US2015/0221740A1, hereinafter Alptekin)**.

6. In reclaim **1**, in FIG. 1-9, Alptekin discloses a semiconductor structure including a contact(**36**) to a semiconductor substrate(**10**), the contact(**36**) comprising:

- a contact opening(**30S&D**) in a dielectric layer(**26P**) over the semiconductor substrate(**10**);
- a first liner layer(**32L**) substantially coating a sidewall the contact opening(**30S&D**);
- a refractory metal layer(**34L**) substantially coating the first liner layer(**32L**) in the contact opening(**30S&D**);
- a metal(**36**) substantially filling the contact opening(**30S&D**); and
- an intermix region(**38S&D**) at an upper surface of the semiconductor

substrate(10) in the contact opening(30S&D), the intermix(38S&D) region having a composition including a material used in the first liner layer(32L), a material used in the refractory metal layer(34L), and a material used in the semiconductor substrate(10) (See § [0031-0078]).

7. In reclaim 2, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the intermix region(38S&D) includes a titanium-based silicide (See § [0065]).

8. In reclaim 3, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the first liner layer(32L) includes titanium (Ti) and the refractory metal layer(34L) includes titanium nitride (TiN) (See § [0078]).

9. In reclaim 4, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the first liner layer(32L) includes nickel platinum (NiPt) and the refractory metal layer(32L) includes a bilayer having a first refractory metal material including titanium (Ti) and a second refractory metal material including titanium (TiN), the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal(36) (See § [0078]).

10. In reclaim 5, in FIG. 1-9, Alptekin discloses a contact of claim 1, wherein the metal(32L) includes tungsten (W) (See § [0078]).

11. In reclaim 6, in FIG. 9, Alptekin discloses a contact of claim 1, wherein the intermix region(38) has a thickness of approximately 3 nanometers to approximately 5 nanometers (See § [0077]).

Conclusion

13. Any inquiry concerning this communication or earlier communications from the examiner should be directed to LAWRENCE-LINH T. NGUYEN at Lawrence-Linh.Nguyen@uspto.gov, or telephone number (571) 272-6267, or fax number (571) 273-6267. The examiner can be reached on M-F (8AM - 4PM EST).

14. If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 272-8300.

15. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at (866) 217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call (800) 786-9199 (in USA or CANADA) or (571) 272-1000.

/Lawrence-Linh T Nguyen/

Examiner, Art Unit 2897

/Fernando L. Toledo/

Supervisory Patent Examiner, Art
Unit 2897

Notice of References Cited	Application/Control No. 15/826,939		Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.	
	Examiner LAWRENCE-LINH T. NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-2015/0221740 A1	08-2015	Alptekin; Emre	H01L29/456	257/288
	B	US-				
	C	US-				
	D	US-				
	E	US-				
	F	US-				
	G	US-				
	H	US-				
	I	US-				
	J	US-				
	K	US-				
	L	US-				
	M	US-				

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

<i>Index of Claims</i> 	Application/Control No. 15826939	Applicant(s)/Patent Under Reexamination ALPTEKIN ET AL.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

✓	Rejected	-	Cancelled	N	Non-Elected	A	Appeal
=	Allowed	÷	Restricted	I	Interference	O	Objected

☐ Claims renumbered in the same order as presented by applicant		☐ CPA		☐ T.D.		☐ R.1.47			
CLAIM		DATE							
Final	Original	08/22/2018							
	1	✓							
	2	✓							
	3	✓							
	4	✓							
	5	✓							
	6	✓							

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Emre Alptekin

Conf. No.: 6013

Serial No.: 15/826,939

Art Unit: 2897

Filed: November 30, 2017

Examiner: Nguyen, Lawrence-Lin T.

Docket No.: FIS920150066US2
(GFFT-0376-DIV)

Title: **CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME**

Mail Stop Amendment
Commissioner for Patents
P.O. Box 1450
Alexandria, VA 22313-1450

AMENDMENT

This paper is being filed in response to the Office Action dated August 27, 2018. Please amend the above-referenced patent application as follows:

AMENDMENTS TO THE CLAIMS

The following listing of the claims replaces all prior versions, and listings, of claims in the application:

1. (Currently Amended) A semiconductor structure including a contact to a semiconductor substrate, the contact comprising:

- a contact opening in a dielectric layer over the semiconductor substrate;
- a first liner layer substantially coating a sidewall of the contact opening;
- a refractory metal layer substantially coating the first liner layer in the contact opening;
- a metal substantially filling the contact opening; and
- an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate;

wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.

2. (Original) The contact of claim 1, wherein the intermix region includes a titanium-based silicide.

3. (Original) The contact of claim 1, wherein the first liner layer includes titanium (Ti) and the refractory metal layer includes titanium nitride (TiN).

4. (Currently Amended) The contact of claim 1, wherein the first liner layer includes nickel platinum (NiPt), ~~and the refractory metal layer includes a bilayer having a~~ the first refractory metal material ~~including~~ includes titanium (Ti) and ~~[[a]]~~ the second refractory metal material ~~including~~ includes titanium nitride (TiN); ~~the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.~~

5. (Original) The contact of claim 1, wherein the metal includes tungsten (W).

6. (Original) The contact of claim 1, wherein the intermix region has a thickness of approximately 3 nanometers to approximately 5 nanometers.

REMARKS

Claims Status

Claims 1-6 are pending in this application. Claim 1 is currently amended to include subject matter from claim 4, and finds support in, at least, Figure 9. Claim 4 is currently amended for grammatical purposes and to remove redundant language in light of the amendment to claim 1. Claims 2, 3, 5 and 6 remain as originally presented. No new matter is believed to have been entered.

§ 102(a)(2) Rejection

Claims 1-6 are rejected under 35 U.S.C. 102(a)(2) as allegedly being anticipated by Alptekin (US 2015/0221740A1). Applicant respectfully traverses this rejection as follows.

Independent claim 1 recites, in part, “wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal material substantially separating the first refractory metal material and the metal.”

Alptekin does not disclose second transition metal liner 34L/34p (being equated to Applicant’s refractory metal layer) as having multiple layers as required by Applicant’s claim 1. Alptekin also does not disclose a first material of liner 34L/34p as substantially separating the first transition metal liner 32L/32p (being equated to Applicant’s first liner layer) from a second material of liner 34L/34p as required by Applicant’s claim 1. Moreover, Alptekin does not disclose a second material of liner 34L/34p as substantially separating a first material of liner 34L/34p from the metal of contact 36 as required by Applicant’s claim 1. While Alptekin may

disclose that second transition metal liner 34L/34p may comprise multiple transition metals ([0067]-[0068]), Alptekin does not disclose liner 34L/34p as having multiple layers where each layer separates specified elements. Accordingly, Applicant submits that the Office has misapplied the multiple transition metals of Alptekin's liner 34L/34p to Applicant's multiple layers (*see* rejection of claim 4 in Office Action).

For any one of the above-noted deficiencies of Alptekin and/or the Office's misapplication of Alptekin's teachings to Applicant's claims, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 1.

Specifically regarding Applicant's claim 3, wherein the first liner layer includes Ti and the refractory metal layer includes TiN, Applicant submits that the Office has again misapplied the teachings of Alptekin. Alptekin states, numerous times, that the second transition metal liner 34L/34p comprises at least one second transition metal element which differs from the at least one first transition metal element (*see e.g.*, [0067]-[0068]). As such, Alptekin cannot read on Applicant's claim 3 since the TiN of the refractory metal layer does not comprise at least one transition metal element which differs from the Ti of the first liner layer (noting that N is not a transition metal element). Accordingly, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 3 for at least this additional reason.

For reasons similar to those just stated with respect to Applicant's claim 3, Alptekin cannot read on Applicant's claim 4 which requires the first liner layer to include NiPt and the refractory metal layer to include Ti/TiN. Accordingly, Applicant submits that Alptekin fails to disclose or suggest Applicant's claim 4 as well.

In light of the foregoing, Applicant respectfully requests withdrawal of the anticipation rejection over Alptekin.

Also in light of the foregoing, and considering that independent claim 1 has merely been amended to include subject matter from original claim 4, Applicant requests that a next Office Action, should one be issued, be non-final since Applicant's amendments to the claims would not have necessitated a new ground of rejection.

CONCLUSION

Applicant respectfully submits that the Application as presented is in condition for allowance. Should the Examiner believe that anything further is necessary in order to place the application in better condition for allowance, the Examiner is requested to contact the Applicant's undersigned representative at the telephone number listed below.

Respectfully submitted,

/Justine M. Wilbur/
Justine M. Wilbur
Reg. No. 59,678

Date: November 27, 2018

Hoffman Warnick LLC
540 Broadway, 4th Floor
Albany, New York 12207
Phone: (518) 449-0044
Fax: (518) 449-0047

Electronic Acknowledgement Receipt

EFS ID:	34412720
Application Number:	15826939
International Application Number:	
Confirmation Number:	6013
Title of Invention:	CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME
First Named Inventor/Applicant Name:	Emre Alptekin
Customer Number:	133964
Filer:	Justine M. Wilbur
Filer Authorized By:	
Attorney Docket Number:	FIS920150066US2
Receipt Date:	27-NOV-2018
Filing Date:	30-NOV-2017
Time Stamp:	18:20:14
Application Type:	Utility under 35 USC 111(a)

Payment information:

Submitted with Payment	no
------------------------	----

File Listing:

Document Number	Document Description	File Name	File Size(Bytes)/ Message Digest	Multi Part /.zip	Pages (if appl.)
1		FIS920150066US2_Amendmen tToOfficeAction1_27NOV2018_ .pdf	109784 f45c75d7a9e74ab247c3428c4d608b8185d e20f9	yes	6

Multipart Description/PDF files in .zip description			
Document Description		Start	End
Amendment/Req. Reconsideration-After Non-Final Reject		1	1
Claims		2	3
Applicant Arguments/Remarks Made in an Amendment		4	6

Warnings:

Information:

Total Files Size (in bytes):	109784
-------------------------------------	--------

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111
If a new application is being filed and the application includes the necessary components for a filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application.

National Stage of an International Application under 35 U.S.C. 371
If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office
If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875				Application or Docket Number 15/826,939		Filing Date 11/30/2017		☐ To be Mailed	
---	--	--	--	---	--	----------------------------------	--	---------------------------------------	--

ENTITY: ☒ **LARGE** ☐ **SMALL** ☐ **MICRO**

APPLICATION AS FILED – PART I

FOR	NUMBER FILED	NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (j), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(j))	minus 20 =	*	X \$ =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	X \$ =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))			If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).	
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED – PART II

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	11/27/2018	CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	* 6	Minus	** 20	= 0	X \$100 =	0
	Independent (37 CFR 1.16(h))	* 1	Minus	***3	= 0	X \$460 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0

AMENDMENT	(Column 1)	(Column 2)	(Column 3)	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
		CLAIMS REMAINING AFTER AMENDMENT					
	Total (37 CFR 1.16(i))	*	Minus	**	=	X \$ =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	X \$ =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.
 ** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".
 *** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".
 The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.

HSLIE
DORIS BURNS

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



NOTICE OF ALLOWANCE AND FEE(S) DUE

133964 7590 03/23/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

EXAMINER

TOLEDO, FERNANDO L

ART UNIT

PAPER NUMBER

2897

DATE MAILED: 03/23/2020

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	06/23/2020

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 1/2 the amount of undiscounted fees, and micro entity fees are 1/2 the amount of small entity fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed and an extra copy of the form should be submitted. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via EFS-Web.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence including the Patent, advance orders and notification of maintenance fees will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications.

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

133964 7590 03/23/2020
HOFFMAN WARNICK LLC
540 BROADWAY
4TH FLOOR
ALBANY, NY 12207

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via EFS-Web or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013

TITLE OF INVENTION: CONTACTS TO SEMICONDUCTOR SUBSTRATE AND METHODS OF FORMING SAME

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1000	\$0.00	\$0.00	\$1000	06/23/2020

EXAMINER	ART UNIT	CLASS-SUBCLASS
TOLEDO, FERNANDO L	2897	257-757000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/SB/47; Rev 03-09 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required) ☐ Advance Order - # of Copies _____

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via EFS-Web ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
15/826,939	11/30/2017	Emre Alptekin	FIS920150066US2	6013
133964	7590	03/23/2020	EXAMINER	
HOFFMAN WARNICK LLC			TOLEDO, FERNANDO L	
540 BROADWAY			ART UNIT	
4TH FLOOR			PAPER NUMBER	
ALBANY, NY 12207			2897	

DATE MAILED: 03/23/2020

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b) (2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Notice of Allowability	Application No. 15/826,939	Applicant(s) Alptekin et al.	
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to 11/27/2018.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

3. ☒ The allowed claim(s) is/are 1-6. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☐ All b) ☐ Some *c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.
2. ☐ Certified copies of the priority documents have been received in Application No. _____.
3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.
THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.

Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).

6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☒ Notice of References Cited (PTO-892) 2. ☐ Information Disclosure Statements (PTO/SB/08), Paper No./Mail Date _____. 3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material _____. 4. ☐ Interview Summary (PTO-413), Paper No./Mail Date _____.	5. ☐ Examiner's Amendment/Comment 6. ☒ Examiner's Statement of Reasons for Allowance 7. ☐ Other _____.
--	--

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897	/FERNANDO L TOLEDO/ Supervisory Patent Examiner, Art Unit 2897
--	---

DETAIL ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Allowable Subject Matter

2. The following is an examiner's statement of reasons for allowance
 - a. With respect to independent claim 1, the prior art, Alptekin et al., (US2015/0221740A1) teaches:
 - 1. (Currently Amended) A semiconductor structure including a contact to a semiconductor substrate, the contact comprising: a contact opening in a dielectric layer over the semiconductor substrate; a first liner layer substantially coating a sidewall of the contact opening; a refractory metal layer substantially coating the first liner layer in the contact opening; a metal substantially filling the contact opening; and an intermix region at an upper surface of the semiconductor substrate in the contact opening, the intermix region having a composition including a material used in the first liner layer, a material used in the refractory metal layer, and a material used in the semiconductor substrate.
 - But the prior arts of record do not teach or fairly suggest a limitation of: wherein the refractory metal layer includes a multi-layer having at least a first refractory metal material and a second refractory metal material, the first refractory metal material substantially separating the first liner layer and the second refractory metal material, the second refractory metal

material substantially separating the first refractory metal material and the metal.

- *Since Unreacted (non-diffused) portions of the first liner layer may be removed prior to completion of fabricating the contact. Therefore, aspects of the present invention do not increase the deposited metal thickness in the contact openings. Thus, contact opening fill issues are avoided and overall contact resistance is improved.* There is no reason apparent to the examiner that all the features claimed in claim **1** would be obvious over the prior art.

- Dependent claims **2-6** are respectively dependent on currently amended allowable independent claim **1**. Therefore, claims **2-6** incorporate the allowable limitations of claim **1**. Consequently, claims **2-6** are allowable.

3. Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance".

Conclusion

4. Any inquiry concerning this communication or earlier communications from the examiner should be directed to Lawrence-Linh Nguyen (Email @ Lawrence-Linh.Nguyen@uspto.gov) whose telephone number is (571) 272-6267 and fax number is (571) 273-6267. The examiner can normally be reached on Monday-Friday (Flex hours 10A-6P). If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Fernando Toledo can be reached on (571) 272-1867. The fax phone number for the organization where this application or proceeding is assigned is (571) 273-8300.

5. Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/LAWRENCE-LINH T NGUYEN/

Examiner, Art Unit 2897

/FERNANDO L TOLEDO/

Supervisory Patent Examiner, Art Unit 2897

Notice of References Cited	Application/Control No. 15/826,939		Applicant(s)/Patent Under Reexamination Alptekin et al.	
	Examiner LAWRENCE-LINH T NGUYEN		Art Unit 2897	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Name	CPC Classification	US Classification
*	A	US-20150221740-A1	08-2015	Alptekin; Emre	H01L29/456	257/288
	B					
	C					
	D					
	E					
	F					
	G					
	H					
	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date MM-YYYY	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title Date, Publisher, Edition or Volume, Pertinent Pages)
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in MM-YYYY format are publication dates. Classifications may be US or foreign.

Search Notes 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC - Searched*		
Symbol	Date	Examiner

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Each of the listed CPC, inventor(s) and assignees are searched in combination and/or delimited with text. See EAST search for details.	12/23/2018	

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	H01L21/7684 H01L21/76843 H01L21/76846 H01L21/76879 H01L21/76883 H01L23/5283 H01L23/53266 H01L23/53261	12/23/2018	

--	--

Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

CPC						
Symbol					Type	Version
H01L	/	23	/	53266	F	2013-01-01
H01L	/	21	/	76843	I	2013-01-01
H01L	/	21	/	76846	I	2013-01-01
H01L	/	21	/	76879	I	2013-01-01
H01L	/	21	/	7684	I	2013-01-01
H01L	/	21	/	76883	I	2013-01-01
H01L	/	23	/	485	I	2013-01-01
H01L	/	21	/	28518	I	2013-01-01
H01L	/	21	/	76855	I	2013-01-01
H01L	/	23	/	53223	A	2013-01-01
H01L	/	23	/	53238	A	2013-01-01
H01L	/	21	/	2855	A	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

INTERNATIONAL CLASSIFICATION			
CLAIMED			
H01L	/	23	/ 532
H01L	/	21	/ 768
NON-CLAIMED			
	/		/

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9

Issue Classification 	Application/Control No. 15/826,939	Applicant(s)/Patent Under Reexamination Alptekin et al.
	Examiner LAWRENCE-LINH T NGUYEN	Art Unit 2897

☒ Claims renumbered in the same order as presented by applicant ☐ CPA ☐ T.D. ☐ R.1.47															
CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1														
	2														
	3														
	4														
	5														
	6														

/LAWRENCE-LIN T NGUYEN/ Examiner, Art Unit 2897 (Assistant Examiner)	25 December 2018 (Date)	Total Claims Allowed: 6	
/FERNANDO L TOLEDO/ SPE, Art Unit 2897 (Primary Examiner)	26 December 2018 (Date)	O.G. Print Claim(s) 1	O.G. Print Figure 9